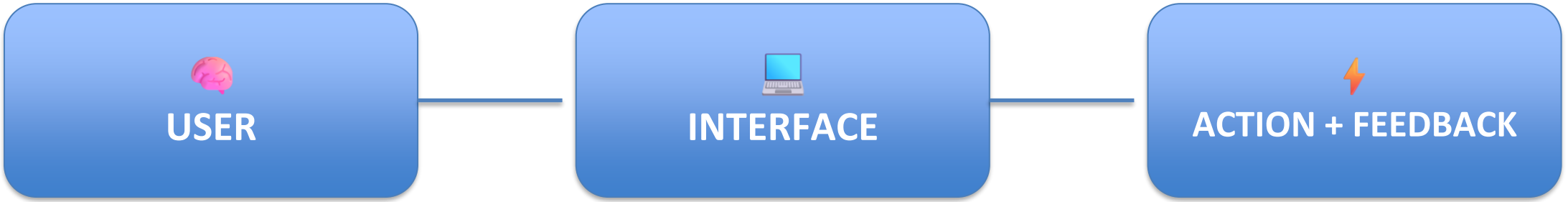
Mental Models & Memory in HCI

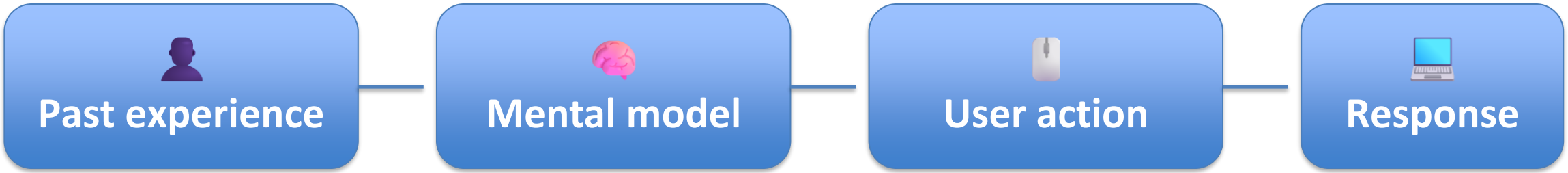
A visual, fun and exam-friendly guide



Big idea: Good HCI matches what users expect and reduces unnecessary memory effort.

1. What is a Mental Model?

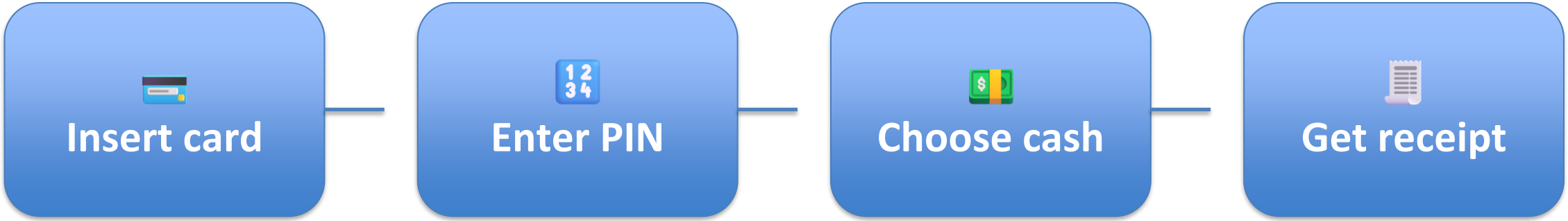
The user's internal idea of how a system works



Example: 🛒 Shopping cart
You already know a real basket holds things you want to buy. So you expect “Add to Cart” to save an item for later checkout.

Example: ATM 🏧

Mental models help users predict the next step



If the ATM suddenly asks for a PIN before inserting the card, the user's mental model is challenged.

Why Mental Models Matter in HCI

🔮 **Predictability**

Users can anticipate outcomes.

📚 **Learnability**

Familiar ideas are easier to learn.

❌ **Fewer errors**

Clear mappings reduce mistakes.

💬 **Feedback**

Responses help users update their model.

🔁 **Consistency**

Same controls should behave similarly.

2. Memory in HCI

Think of memory as a funnel: see → hold → store → retrieve



HCI goal: Don't make users remember what the interface could simply show.

Sensory Memory 👀 👂 ✋

The first, extremely brief stage of memory

👁️ ICONIC
Visual information

Example: a flashing icon

👂 ECHOIC
Auditory information

Example: remembering a spoken word

✋ HAPTIC
Touch information

Example: phone vibration

Quick idea: Sensory memory is like a tiny “buffer” before attention decides what matters.

Short-Term & Working Memory 🧩

Temporary information used during the current task

🔢 **OTP: 482731**
Remember it while typing

🧮
Hold numbers while calculating

🧭
Remember the last step while doing the next

Design tip 💡 : Keep important information visible. If users must switch screens just to remember a detail, working-memory load increases.

Long-Term Memory 🗄️

Information that can remain available for a long time

📖
SEMANTIC
Facts & meanings

“Trash icon = delete”

📸
EPISODIC
Personal events

“My last online order”

⌨️
PROCEDURAL
Skills / how-to

Typing & shortcuts

🔄
IMPLICIT
Past experience

Familiar patterns feel easy

Semantic vs. Episodic Memory

A simple way to remember the difference

📚 **SEMANTIC**
“What do I KNOW?”

Example: The floppy-disk icon often means Save.

📸 **EPISODIC**
“What do I REMEMBER EXPERIENCING?”

Example: I remember saving my assignment yesterday.

Memory trick: Semantic = facts. Episodic = episodes.

Recognition vs. Recall 👀 🧠

A major HCI idea

👀 **RECOGNITION**
See the choice → identify it

Example: Clicking a visible “Settings” button

🧠 **RECALL**
Retrieve it without seeing it

Example: Remembering a hidden keyboard shortcut

HCI generally tries to support recognition when practical.

Designing for Human Memory 🎨

Make the interface do more of the remembering

- 👁️ Show important information instead of hiding it.
- 🏷️ Use meaningful labels and familiar icons.
- 🔁 Keep interactions consistent.
- 💬 Give clear feedback after actions.
- 🧭 Provide cues, suggestions, and navigation.
- 🧠 Avoid unnecessary memory-heavy steps.

Revision 🚀

Remember this for your HCI exam

🧠 MENTAL MODEL

User's expectation of how a system works.

Example: shopping cart, ATM

💾 MEMORY

Sensory → brief
Working → temporary
Long-term → durable

🗂️ Long-TERM

Semantic = facts
Episodic = events
Procedural = skills
Implicit = influence of past experience

Golden rule: Design for recognition, consistency, feedback, and low memory load.